

# Clinical Risk Assessment Report: Patient 47

## 1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Borderline / Low Confidence
- Absolute Predicted Risk: 17.4% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.79x the diagnostic threshold

### Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Tumor Sphericity (CT) (SHAPE\_Sphericity(onlyFor3DROI)).CT), which reduced the patient's absolute risk by 1.6%. Biologically, this feature quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

## 2. Key Pathological & Imaging Drivers (Elevating Risk)

**Squamous Cell Carcinoma Antigen (SCC\_Ag)** **Value: 0.5 [Bottom 10%] (+2.3%)**  
*Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.*

**Small Zone Emphasis (PET) (GLZLM\_SZE.PET)** **Value: 0.18 [Avg (58th %ile)] (+2.0%)**  
*Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.*

**Tumor Pixel Correlation (CT) (GLCM\_Correlation.CT)** **Value: -0.72 [Bottom 22%] (+1.3%)**  
*Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.*

## 3. Protective / Mitigating Factors (Reducing Risk)

**Tumor Sphericity (CT) (SHAPE\_Sphericity(onlyFor3DROI)).CT)** **Value: 0.58 [Avg (72th %ile)] (-1.6%)**  
*Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.*

**Large Zone Low Gray-Level Emphasis (CT) (GLZLM\_LZLGE.CT)** **Value: -0.37 [Bottom <1%] (-1.2%)**  
*Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.*

**Carcinoembryonic Antigen (CEA)** **Value: 1.71 [Bottom 20%] (-1.0%)**  
*Elevated levels are strongly associated with increased tumor burden and potential for micrometastasis.*

**Short Run High Gray-Level Emphasis (CT) (GLRLM\_SRHGE.CT)** **Value: 0.28 [Avg (52th %ile)] (-0.9%)**  
*Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.*

**75th Percentile Tumor Density (CONVENTIONAL\_HUQ3)** **Value: 0.43 [Avg (56th %ile)] (-0.8%)**  
*Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.*

Machine Learning Feature Attribution (SHAP Decision Path)

